

ULIP Integration Requirement Document

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Revision History

Version	Date	Author	Reviewer	Significant Changes
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Introduction

1.1 Purpose

The purpose of this document is to define and provide details for accessing API for E-Challan system information (information about vehicle number available in E-Challan system) from ULIP system.

This document details the following points:

1. **Data Integration between E-Challan and ULIP web service:** Identification of operational points where data integration between **E-Challan** and ULIP web service will be needed.
2. **Technical Approach** for performing Data Integration

The Overall Description

User will share the information such as **vehicle number** based on data send by user ULIP system will call respective **E-Challan** API for accessing information.

Following are the point for accessing vehicle information: -

1. This API will share data based on information provided by user.
2. ULIP will expose API for getting data from E-Challan API.
3. **ECHALLAN/01** API will take vehicle number from user and get data from E-Challan API.

1.2 Data Integration between E-Challan System and ULIP application

User will share information to ULIP system based on data shared ULIP will provide data to user.

1.3 ECHALLAN/01

This API will take **vehicle number** and connect with E-Challan API to get data.

S. No.	Field Name	Field Description	Format	Required/Not – Required	Length
1	vehicleNumber	Enter the vehicleNumber as available in E-Challan system.	^[a-zA-Z0-9]{5,11}\$	Required	Min 5 Max 11

1.3.1 Technical Approach

1. In this, on staging server we are fetching the data from 3rd party E-Challan API directly without hitting any 3rd party AUTH API.
2. In production server for get the data from 3rd party E-Challan API, we have to first hit the 3rd party AUTH API for get the token and then we use the token in the 3rd party E-Challan API, we are using the AUTH token API internally.
3. All the data will be shared through rest web services.
4. ULIP system will provide **vehicle number** for which detail is required.
5. E-Challan API will share details of vehicle details which exist in E-Challan system in JSON format.
6. ECHALLAN Integration will add JSON in **response** key in response body.
7. ULIP system will provide the data in JSON Object format in response body.

Example:

Vehicle number **KL150000** is shared by ULIP system, then ECHALLA-integration will make a request to E-Challan API.

Request:

The request must be of the following format, where vehicle number is passed within the URL body in JSON format.

For staging -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/ECHALLAN/01>

```
{  
  "vehicleNumber": "UP93BK9110"  
}
```

Curl URL

```
curl --location --  
request POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/ECHALLAN/01' \  
--  
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJlbGwIiwiaWF0IjoxNjk1Mzc1OTgyLCJhcHBzIjoizGF0YXB1c2gifQ.2bntkTWUW7mIdSW2NklLa0cE7hOPwI37uS9FUcf3NDjAnMV31_5SIO  
Ni50704RF39kYM9rVJ59KB2gh7gSheDg' \  
--header 'Content-Type: application/json' \  
--header 'Accept: application/json' \  
--header 'Cookie: __zlcmid=1DSlSDwigKpTJa2; _fw_crm_v=b6a11a26-11c3-4b74-d0d1-3b9c43a5b7c7' \  
--data-raw '{  
  "vehicleNumber": "UP93BK9110"  
}'
```

For production -
<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/ECHALLAN/01>

```
{
  "vehicleNumber": "UP15AU1291"
}
```

Curl URL

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/ECHALLAN/01' \
--header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJlbGlwIiwiaWF0IjoxNzI3NzYyNzcxLCJhcHBzIjoizGF0YXB1c2gifQ.9_kJwxuXfJYOMt0QeMgOSWi01EaxXQkZQ1JLevfNo-vJ5vjCY9oIemZ_B4hiiq2ZCog-yAbIUgL6YN-TVie3cA' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data '{
  "vehicleNumber": "UP15AU1291"
}'
```

For obtaining authentication token, refer [Section 1.4](#)

Response

To acknowledge the above request, ULIP will send the below response -

1. Invalid vehicle number format

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered at: nhno. Format should follow [A-Z]{2}-[A-Z0-9]{1,4}"
}
```

2. In case vehicle number exist

```
{
  "response": [
    {
      "response": {
        "data": {
          "Disposed_data": [
            {
              "challan_place": "",
              "sent_to_reg_court": "No",
              "owner_name": "T**T P*T L*D",
            }
          ]
        }
      }
    }
  ]
}
```

```
"receipt_no": null,
"challan_date_time": "11-03-2022 11:38:21",
"document_impounded": "",
"received_amount": 1,
"sent_to_court_on": null,
"remark": "NA",
"fine_imposed": "1",
"amount_of_fine_imposed": null,
"challan_no": "KL48648220311113821",
"dl_no": null,
"name_of_violator": "T**T P*T L*D",
"sent_to_virtual_court": "No",
"date_of_proceeding": null,
"driver_name": "T**T P*T L*D",
"court_address": null,
"challan_status": "Disposed",
"department": "Traffic",
"state_code": "KL",
"court_name": null,
"offence_details": [
  {
    "act": "",
    "name": "test offence 1 rupee"
  }
]
},
{
  "challan_place": "",
  "sent_to_reg_court": "No",
  "owner_name": "T**T P*T L*D",
  "receipt_no": null,
  "challan_date_time": "25-02-2022 11:20:01",
  "document_impounded": "",
  "received_amount": 1,
  "sent_to_court_on": null,
  "remark": "NA",
  "fine_imposed": "1",
  "amount_of_fine_imposed": null,
  "challan_no": "KL48648220225112001",
  "dl_no": null,
  "name_of_violator": "T**T P*T L*D",
  "sent_to_virtual_court": "No",
  "date_of_proceeding": null,
  "driver_name": "T**T P*T L*D",
  "court_address": null,
  "challan_status": "Disposed",
```

```

        "department": "Traffic",
        "state_code": "KL",
        "court_name": null,
        "offence_details": [
            {
                "act": "",
                "name": "test offence 1 rupee"
            }
        ]
    },
    "Pending_data": [
        {
            "challan_place": "test",
            "sent_to_reg_court": "No",
            "owner_name": "T*T P*T L*D",
            "challan_date_time": "01-09-2023 17:36:00",
            "document_impounded": "",
            "sent_to_court_on": null,
            "remark": "gdgg",
            "fine_imposed": "7750",
            "amount_of_fine_imposed": null,
            "challan_no": "KL548476230713105383",
            "dl_no": null,
            "name_of_violator": "T*T P*T L*D",
            "sent_to_virtual_court": "No",
            "date_of_proceeding": null,
            "driver_name": null,
            "court_address": null,
            "rto_distric_name": "Thiruvananthapuram",
            "challan_status": "Pending",
            "department": "Traffic",
            "state_code": "KL",
            "court_name": null,
            "offence_details": [
                {
                    "act": "S130(4)",
                    "name": "Fitness certificate (CF) of a transport vehicle not produced on demand for examination by the officer authorised."
                },
                {
                    "act": "S66(1),S84(a) RW S192A(1)",
                    "name": "Driving or causing or allowing to be driven a vehicle as contract carriage without valid permit.(MMV and HMV)"
                }
            ]
        }
    ]

```



```
    },
    {
      "challan_place": "",
      "sent_to_reg_court": "No",
      "owner_name": "T**T P*T L*D",
      "challan_date_time": "11-03-2022 11:49:37",
      "document_impounded": "",
      "sent_to_court_on": null,
      "remark": "NA",
      "fine_imposed": "1",
      "amount_of_fine_imposed": null,
      "challan_no": "KL48648220311114937",
      "dl_no": null,
      "name_of_violator": "T**T P*T L*D",
      "sent_to_virtual_court": "No",
      "date_of_proceeding": null,
      "driver_name": "T**T P*T L*D",
      "court_address": null,
      "rto_distric_name": "Palakkad",
      "challan_status": "Pending",
      "department": "Traffic",
      "state_code": "KL",
      "court_name": null,
      "offence_details": [
        {
          "act": "",
          "name": "test offence 1 rupee"
        }
      ]
    }
  ],
  "message": "Record finds successfully",
  "status": "200"
},
"responseStatus": "SUCCESS"
}
],
"error": "false",
"code": "200",
"message": "Success"
}
```

3. In case vehicle number not exist

```
{
  "response": [
    {
      "response": {
        "code": "305",
        "message": "No Records Found!"
      },
      "responseStatus": "SUCCESS"
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

Status Code 200: ECHALLAN data response

Status Code 400: Bad request (Invalid json syntax, invalid json key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server error occurred

Status Code 502: Server is not responding.

1.4 Data Transmission & Authentication Mechanism

All data exchange would be done over secure HTTP (HTTPS). Request from ULIP's system would be made in ECHALLAN system through a data exchange URL that has following form:

For staging -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/ECHALLAN/01>

```
{
  "vehicleNumber": "KL150000"
}
```

Access to above data exchange URL would be authorized by use of a security mechanism implemented by the ULIP system. It is as follows:

- First time access: It would require a username and password [which would be shared with user beforehand]. Once user hit with valid user id and password, ULIP system will return a basic authorization token to user, using that token user will communicate with ULIP system until token will not get expired.

Brief summary of HTTP Basic Authentication is as follows:

First time when User will hit ULIP secured API through username and password for access token as follow-

```
curl --location --request POST
'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/user/login' \
  --header 'Accept: application/json' \
  --header 'Content-Type: application/json' \
  --data-raw '{
    "username": "xxxx",
    "password": "xxxx@123"
  }'
```

Authorization:

Bearer

eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGlwliwiaWF0IjoxNjc1Mzc5NzlyLCJhcHBzljoiZGF0YXB1c2gifQ.d8gwtG3TkAPKKk4Gvz8Moi6nCtU3wQvZnJOA8eG2e2afY4TsTliy6uQERhFbD30WliHlsTRo3LldgUctEa5Bdw'

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired and user need to re authenticate.

Using The above token, user can request ULIP system for cargo details as follow –

```
curl --location --
request POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/ECHALLAN/01' \
--
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGlwliwiaWF0IjoxNjc1Mzc5NzlyLCJhcHBzljoiZGF0YXB1c2gifQ.d8gwtG3TkAPKKk4Gvz8Moi6nCtU3wQvZnJOA8eG2e2afY4TsTliy6uQERhFbD30WliHlsTRo3LldgUctEa5Bdw' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--header 'Cookie: __zlcmid=1DSISDwigKpTJa2; _fw_crm_v=b6a11a26-11c3-4b74-d0d1-3b9c43a5b7c7' \
--data-raw ' {
  "vehicleNumber": "UP93BK9110"
}'
```

For production –

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/user/login' \  
--header 'Accept: application/json' \  
--header 'Content-Type: application/json' \  
--data-raw '{  
  "username": "xxxx",  
  "password": "xxxx@123"  
'
```

Authorization:**Bearer**

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGllwIiwiaWF0IjoxNzI3MTYxMDgwLCJhcHBzIjoizGF0YXB1c2gifQ.XhVZm7NiCD05-v9091WJQhYzdk1a0Shs_Ley0T4R2MX63swRyxHbHxh6bVXTzqWvWiSLwwNGRQIcqGhr49hpGg
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired and user need to re authenticate.

Using The above token, user can request ULIP system for details as follow -

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/ECHALLAN/01>

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/ECHALLAN/01' \  
--header 'Authorization: Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGllwIiwiaWF0IjoxNzI3MTYxMDgwLCJhcHBzIjoizGF0YXB1c2gifQ.9_kJwxuXfJYoMt0QeMgOSWi01EaxXQkZQ1JLevfNo-vJ5vjCY9olemZ_B4hiiq2ZCog-yAbIUg6YN-TVie3cA' \  
--header 'Content-Type: application/json' \  
--header 'Accept: application/json' \  
--data '{  
  "vehicleNumber": "UP15AU1291"  
'
```